AegisLink

Console-Based Encrypted File & Message Transfer System

Software Project Lab

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Project Description

AegisLink enables secure file and message transfer between clients over a TCP network. All data is encrypted before transmission and verified for integrity on arrival.

Encryption	Integrity	Networking
DES and AES algorithms encrypt all messages and files before sending.	SHA-512 hashing detects tampering — server flags [INTACT] or [TAMPERED!].	Multithreaded TCP server on port 314159 handles multiple clients at once.

Motivation

01

Security

Data sent over a network can be intercepted. DES and AES encryption ensures only the intended receiver can read it.

02

Integrity

Data can be altered in transit. SHA-512 hashing lets the server verify every message and file arrived unmodified.

03

Usability

Console-based and lightweight — no complex setup needed for either the client or server.

Working Methodology

01

Connection

Client connects to server via TCP (port 314159), enters name, selects DES or AES.

02

Encryption

Message or file is encrypted locally before leaving the client.

03

Hashing

SHA-512 hash of the ciphertext is computed and attached to the data.

04

Transmission

Encrypted data + hash is sent over the TCP socket to the server.

05

Verification

Server recomputes hash and compares — flags [INTACT] or [TAMPERED!].

06

Relay

If intact, server decrypts and forwards to all other connected clients.

Technology Used

DES — Data Encryption Standard

- 64-bit block, 56-bit key, 16 rounds
- Feistel network structure
- Legacy — included for comparison

AES — Advanced Encryption Standard

- 128-bit block, 128-bit key, 10 rounds
- SubBytes → ShiftRows → MixColumns
- Current industry standard

SHA-512 — Hashing

- Produces 512-bit digest of ciphertext
- 1-bit change → completely different hash
- Flags [INTACT] or [TAMPERED!]

TCP Socket Programming

- Winsock2, port 314159
- std::thread per connected client
- Supports messages and file transfer

How the Algorithms Work

DES

64-bit Plaintext

Initial Permutation

Split: L(32) + R(32)

16 Feistel Rounds

Swap Halves

Final Permutation

64-bit Ciphertext

AES

128-bit Plaintext

AddRoundKey (Key 0)

SubBytes

ShiftRows

MixColumns

AddRoundKey ×10

128-bit Ciphertext

SHA-512

Ciphertext Input

Padding (×1024 bits)

Init H0–H7

80 Mixing Rounds

Add to Hash Values

Compare Hashes

[INTACT] / [TAMPERED!]

Progress

Done

TCP Server Implementation
Multithreaded, handles multiple clients

Done

TCP Client Implementation
Connect, send messages and files

Done

Client-Server Communication
Bidirectional message relay confirmed

Done

DES Algorithm
Encrypt / decrypt messages and files

Done

AES Algorithm
Encrypt / decrypt messages and files

Done

SHA-512 Integrity Verification
Hash check on every transfer

Output — Encrypted Chat (DES)

Client 1 — meem

```
C:\Users\Jannatul Meen
--- ENCRYPTED CHAT (DES & AES) ---
Connected to server!

Enter your name: meem
Choose encryption (1=DES / 2=AES): 1

[Chat ready | meem | DES]
  /file <path> - send a file
  /quit        - exit
----------------------------------------
> halo. ich bin meem.
>
>> nabila : halo. ich heiBe maria. wie geht's
?
> sehr gut. und du?
>
>> nabila : mir geht es gut.
> freut mich.
>
```

Client 2 — nabila

```
C:\Users\Jannatul Meen
--- ENCRYPTED CHAT (DES & AES) ---
Connected to server!

Enter your name: nabila
Choose encryption (1=DES / 2=AES): 1

[Chat ready | nabila | DES]
  /file <path> - send a file
  /quit        - exit
----------------------------------------
>
>> meem : halo. ich bin meem.
> halo. ich heiBe maria. wie geht's?
>
>> meem : sehr gut. und du?
> mir geht es gut.
>
>> meem : freut mich.
>
```

Server

```
C:\Users\Jannatul Meen
   ENCRYPTED SERVER - DES & AES Support
Server started on port: 314159
Supports DES and AES encryption
Waiting for clients...

Client connected.
Client connected.
[INTACT][DES] meem : halo. ich bin meem.
[INTACT][DES] nabila : halo. ich heiBe maria.
 wie geht's?
[INTACT][DES] meem : sehr gut. und du?
[INTACT][DES] nabila : mir geht es gut.
[INTACT][DES] meem : freut mich.
```

Server logs show [INTACT][DES] — confirming messages arrived unmodified.

Output — File Transfer (AES) + Integrity Check

Client — sending file.txt via AES

```
C:\Users\Jannatul Meen
--- ENCRYPTED CHAT (DES & AES) ---
Connected to server!

Enter your name: meem
Choose encryption (1=DES / 2=AES): 2

[Chat ready | meem | AES]
  /file <path> - send a file
  /quit        - exit
----------------------------------------
> /file C:\Users\Jannatul Meem\file.txt
[File sent: file.txt | AES]
>
```

Server — receiving, hash check, save

```
C:\Users\Jannatul Meen
   ENCRYPTED SERVER - DES & AES Support
Server started on port: 314159
Supports DES and AES encryption
Waiting for clients...

Client connected.
[INTACT][DES] meem : quit
Client Disconnected!
Client connected.
Client Disconnected!
Client connected.
Receiving encrypted file: file.txt (272 bytes
) [AES]
[TAMPERED!] File integrity check failed: file
.txt
File "file.txt" saved successfully! [AES]
```

Server detects [TAMPERED!] on hash mismatch, then saves the file successfully with AES decryption.

Thank You

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GitHub Repository

https://github.com/Jfmeem/AegisLink